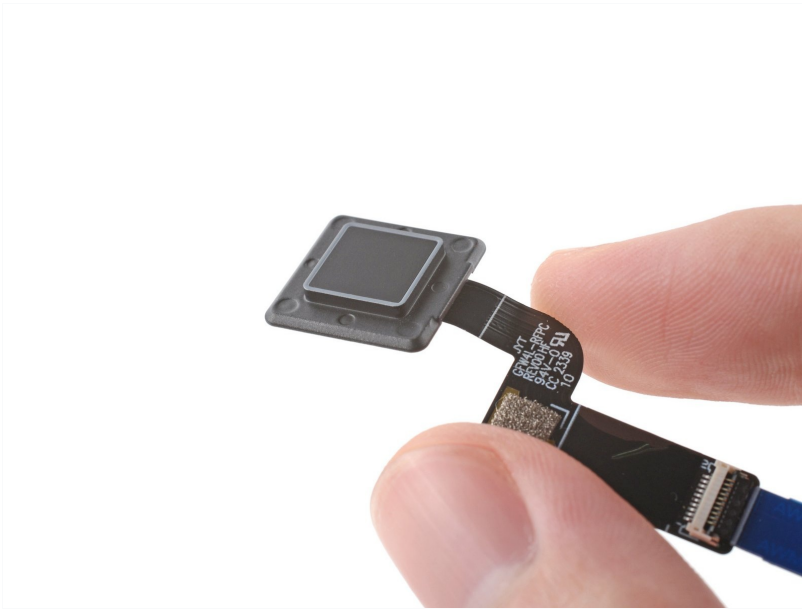




Framework Laptop 16" Fingerprint Sensor Replacement

Use this guide to replace the fingerprint...

Written By: Alex Diaz-Kokaisl



INTRODUCTION

Use this guide to replace the fingerprint sensor in your Framework 16" laptop.

Note: the laptop used in this guide is a generic setup of one US keyboard, one numpad module, and two touchpad spacers. Your setup might vary, but the procedure should remain the same.

TOOLS:

T5 Torx Screwdriver (1)

Phillips #0 Screwdriver (1)

Spudger (1)

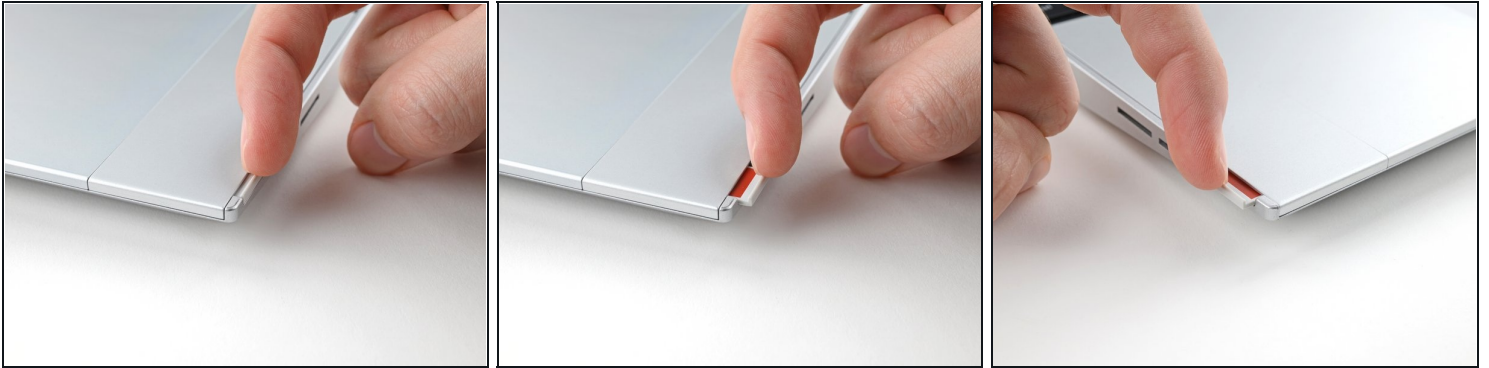
Precut Double-Sided Adhesive Card (1)

Step 1 — Before you begin



- Unplug all cables and fully shut down your laptop.
- ① If your repair requires a spudger, a T5 Torx, or a Phillips screwdriver, you can use the included Framework screwdriver. Whichever works best for you!
- ① Framework recommends that you put your laptop into **battery disconnect mode** in the BIOS whenever removing the battery. Read how to do that [here](#).
- ① To disable this mode, click [here](#).

Step 2 — Unlock the Input Modules



- Use your fingernail to pull out the two Input Module latches and unlock them.
 - ① The latch will show red if it's unlocked.

Step 3 — Remove the touchpad spacers



- Use your fingers to slide the touchpad spacer toward the bottom edge of the laptop and unclip it.
 - ① If you're having trouble, check if the corresponding Input Module latch is properly unlocked.
- Lift the touchpad spacer off the laptop and remove it.

Step 4



- Repeat the same procedure for the other touchpad spacer.

Step 5 — Remove the touchpad



- Use your fingers to slide the touchpad toward the bottom edge of the laptop and disconnect it.
 - ⓘ If you're having trouble, check if the Input Module latches are properly unlocked.
 - Lift the touchpad and remove it.
- ✦ During reassembly, perform the following:
- Place the touchpad flat on its cutout so [its clips](#) are properly aligned.
 - Press the touchpad down and slide it into place so it lines up evenly with the bottom edge of the laptop.

Step 6 — Remove the keyboard



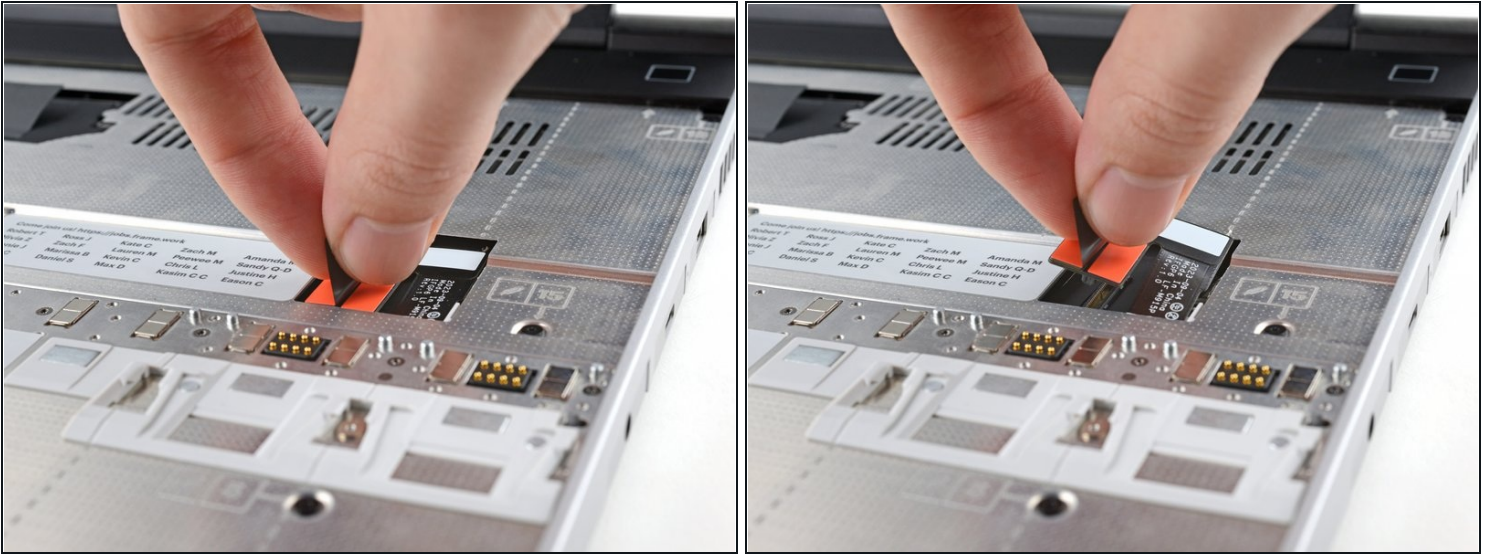
- ① The keyboard is held in place with strong magnets. Apply gradually increasing force to avoid having the keyboard violently eject.
- Grip the two pull tabs along the bottom of the keyboard.
- Lift the pull tabs until the keyboard magnets release.
- Remove the keyboard.
- ★ During reassembly, make sure the keyboard is seated properly on its [alignment pegs](#) and sits evenly with the edges of the laptop.

Step 7 — Remove the numpad



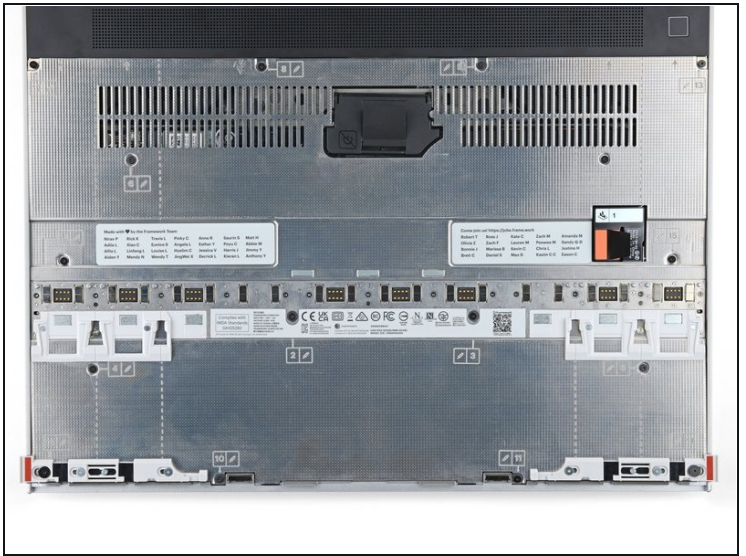
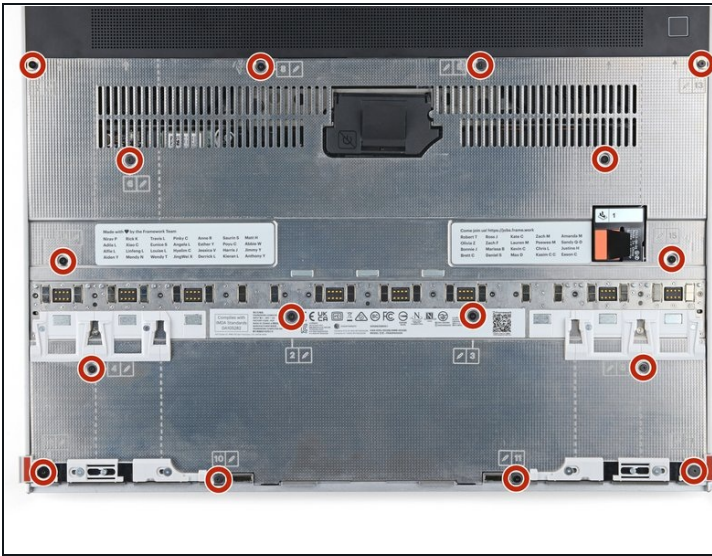
- ① The numpad is held in place with strong magnets. Apply gradually increasing force to avoid having the numpad violently eject.
- Grip the pull tab at the bottom of the numpad.
- Lift the pull tab until the numpad magnets release.
- Remove the numpad.
- ★ During reassembly, make sure the numpad is seated properly on its [alignment pegs](#) and sits flush with the edges of the laptop.

Step 8 — Disconnect the midframe cable



- Grip the black pull tab on the midframe cable press connector.
- Lift up to disconnect the midframe cable.

Step 9 — Loosen the midframe screws



- ① The midframe screws are ordered from 2–17 (number 1 is the press connector). You don't have to follow the order, but you can use it to help keep track of the screws you've loosened.
- Use a T5 Torx screwdriver to loosen the 16 captive screws securing the midframe.
- ☑ During reassembly, **follow the order** (starting with 2) to reinstall the screws across the midframe evenly.

Step 10 — Remove the midframe



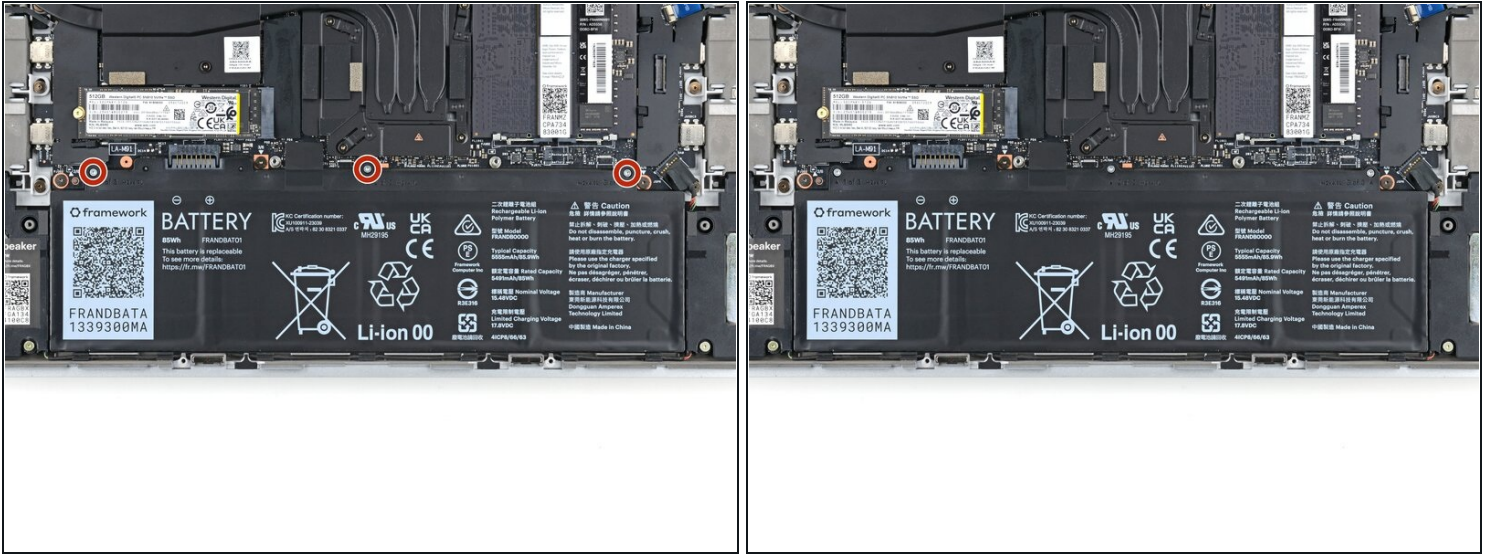
- Use your fingernail to lift the bottom edge of the midframe enough to grip it with your fingers.

Step 11



- Lift the midframe off the laptop and remove it.
- ☑ During reassembly, make sure the midframe sits evenly on its [alignment pegs](#) before tightening the screws.

Step 12 — Loosen the battery screws



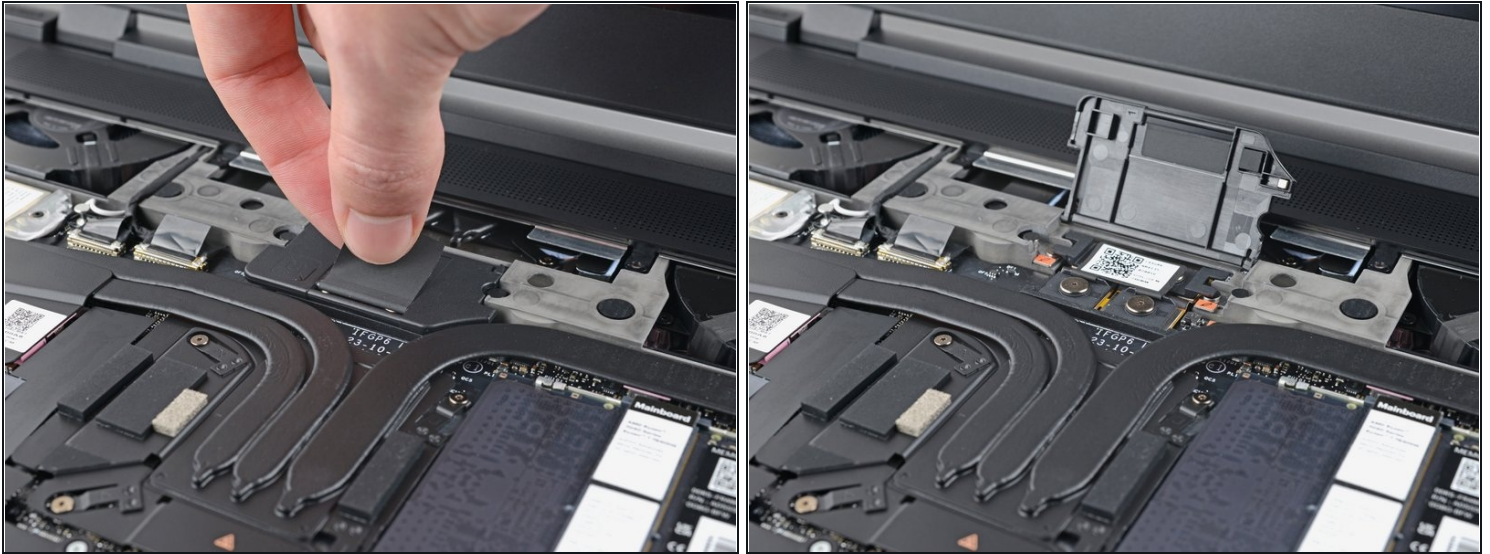
- Use a T5 Torx screwdriver to loosen the three captive screws securing the battery.

Step 13 — Remove the battery



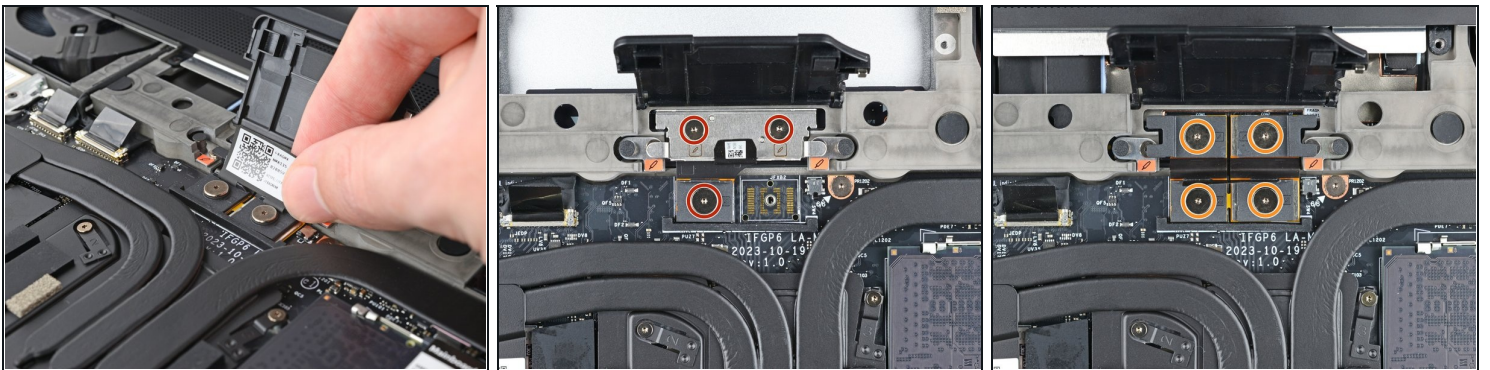
- Grip the black pull tab at the top of the battery.
- Lift the pull tab to disconnect the battery connector.
- Remove the battery.

Step 14 — Lift the interposer door



- Lift the interposer door by its black pull tab and let it rest upright.

Step 15 — Loosen the interposer screws



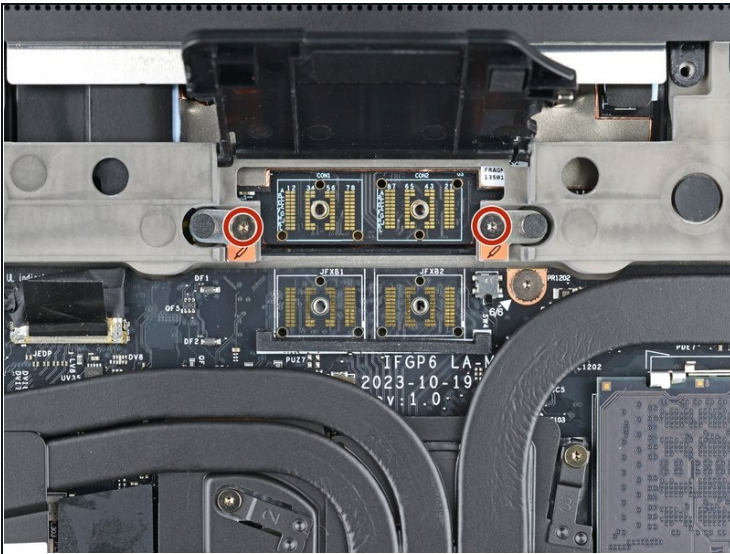
- ① If you have the graphics module installed, your interposer will have four screws. Otherwise, you'll have three screws.
- Lift the pull tab on the interposer and let it rest vertically to access all of its screws.
- Use a T5 Torx screwdriver to loosen the three captive screws securing the interposer.
- If you have the graphics module, use a T5 Torx screwdriver to loosen the four captive screws securing the interposer.

Step 16 — Remove the interposer



- Lift the interposer by its pull tab and remove it.

Step 17 — Loosen the expansion bay screws



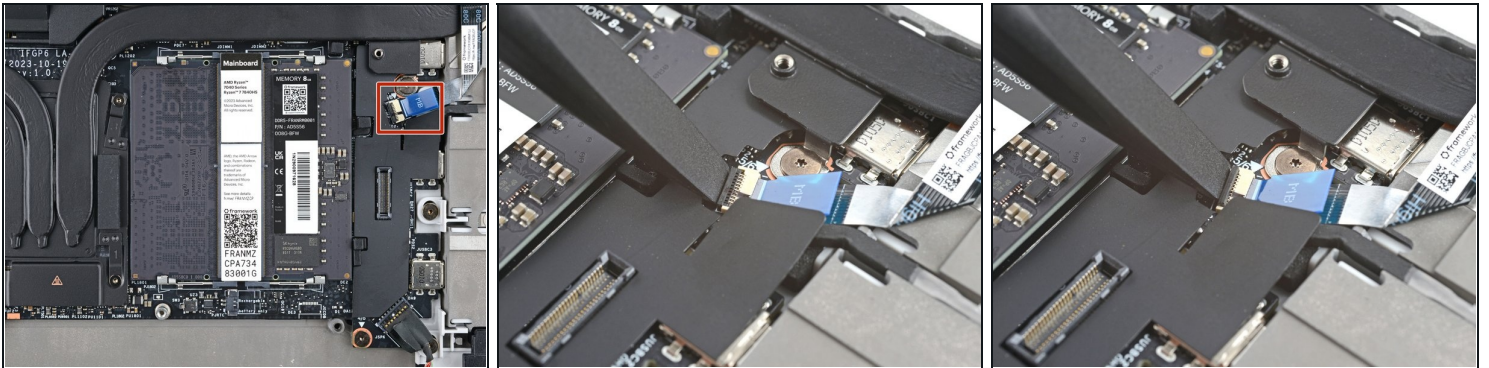
- Use a T5 Torx screwdriver to loosen the two captive screws securing the expansion bay.
- Close the interposer door before continuing.

Step 18 — Remove the Expansion Bay module



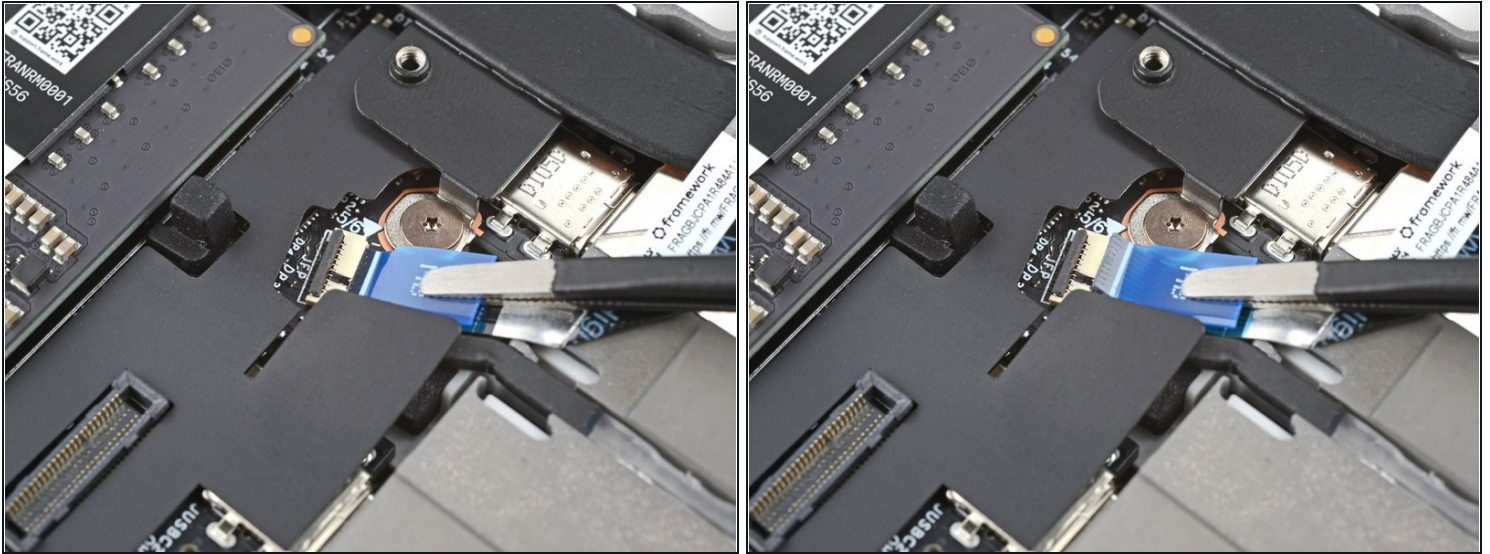
- Slide the Expansion Bay module out of the laptop and remove it.
 - ① The module should slide out easily. If you feel any resistance, check if the screws holding it in place are fully loosened.

Step 19 — Disconnect the fingerprint sensor



- Use the flat end of a spudger, or a clean fingernail, to lift up the locking tab on the fingerprint sensor ZIF connector.

Step 20



- Use [tweezers](#), or your fingers, to grip the blue pull tab and slide the fingerprint sensor cable straight out of its socket.

Step 21



- ❗ The fingerprint sensor cable is lightly adhered to the laptop frame.
- Use your fingers to peel the fingerprint sensor cable away from the frame and separate the adhesive.
- ☑ During reassembly, if your adhesive is still sticky, press the cable to the frame to re-adhere it. Otherwise, use double-sided tape to replace the adhesive.

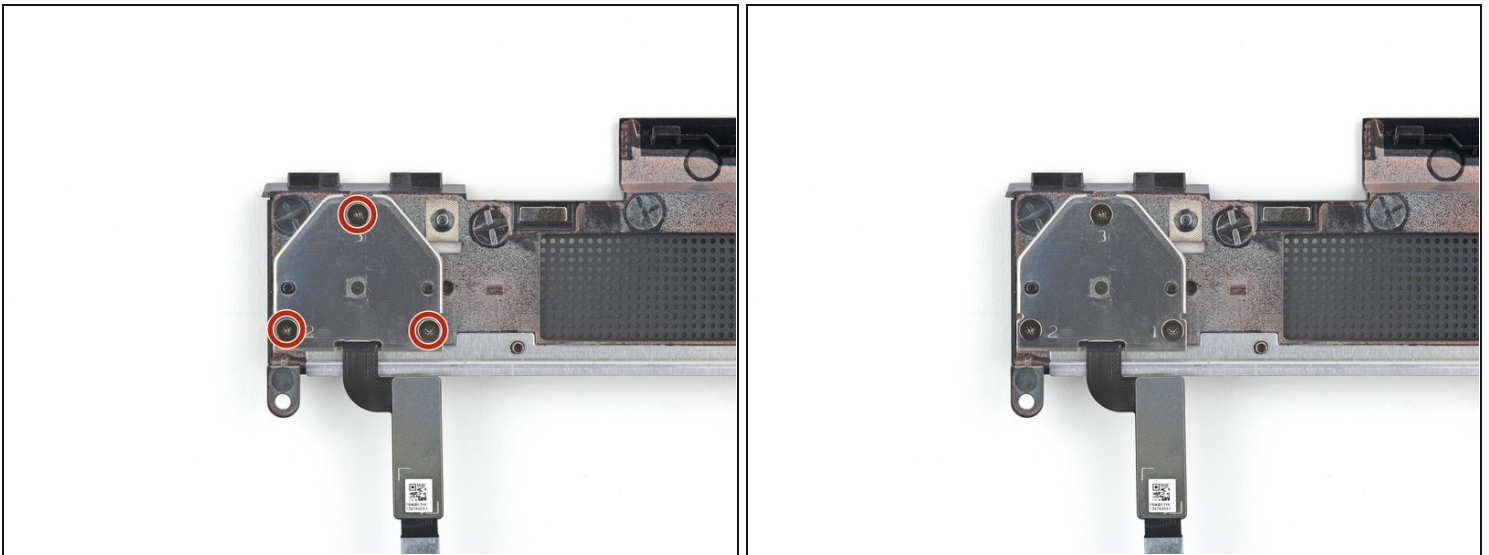
Step 22 — Remove the ventilation plate



ⓘ The ventilation plate is held in place with magnets.

- Lift the bottom of the ventilation plate and pull it away from the laptop until the magnets release.
- Remove the ventilation plate.

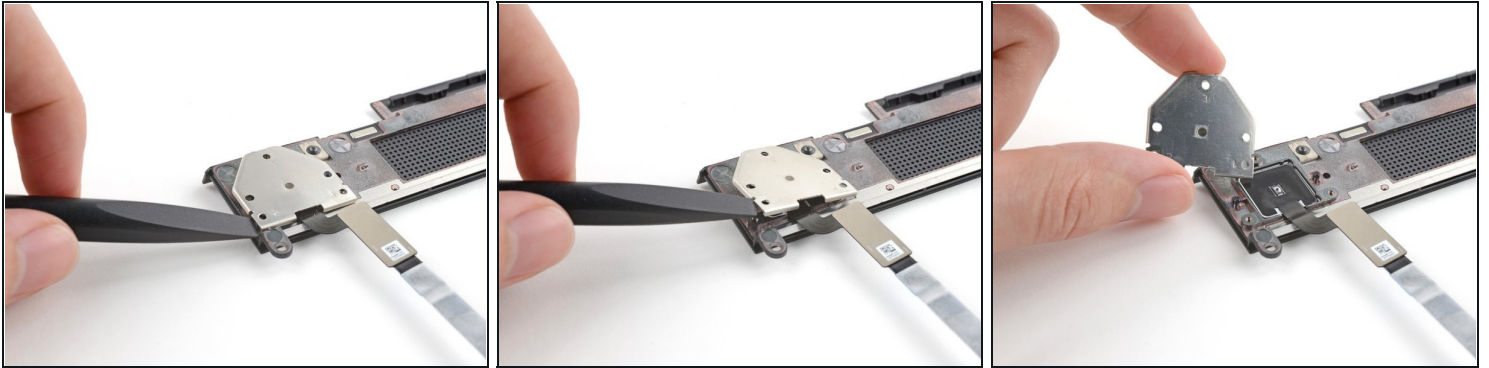
Step 23 — Remove the fingerprint sensor cover screws



ⓘ The fingerprint sensor cover screws are marked 1–3. You don't need to follow this order, but you can use it to help keep track of the screws you've removed.

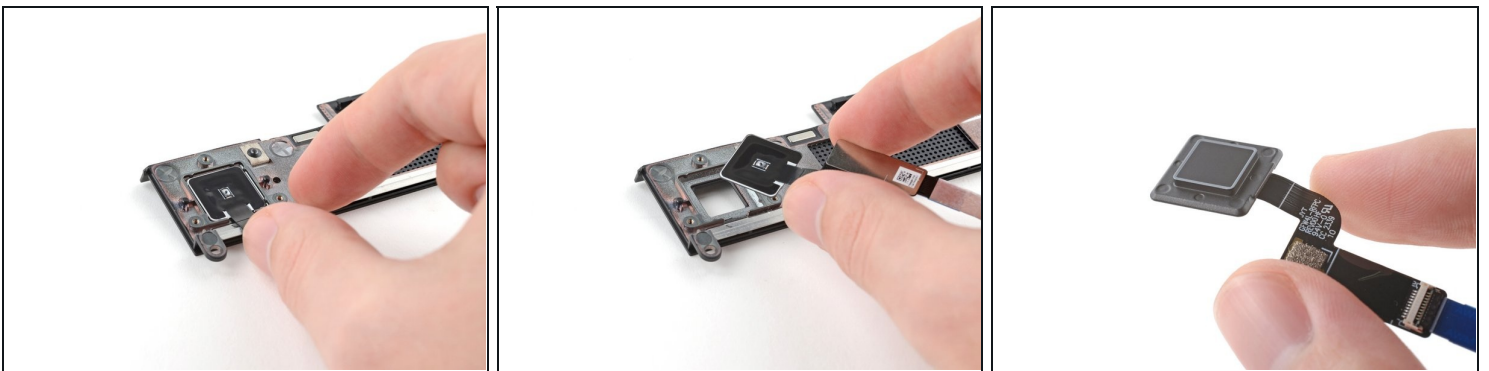
- Use a Phillips screwdriver to remove the three 2 mm-long screws securing the fingerprint sensor cover.

Step 24 — Remove the fingerprint sensor



- Use a spudger, or your fingers, to lift the fingerprint sensor cover and remove it.

Step 25



- Lift the fingerprint sensor out of its slot in the ventilation plate and remove it.

To reassemble your device, follow these instructions in reverse order.

Take your e-waste to an [R2 or e-Stewards certified recycler](#).

Repair didn't go as planned? Try some [basic troubleshooting](#), or ask our [Answers Community](#) for help.